Adaptive Light-Triggered Alaram System Using Op-Amp Comparator Circuit

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Abstract—: The invention described herein relates to an ambient-light-control alarm system utilizing a comparator type of operational amplifier (opamp). An ambient-light level is measured continuously with a light dependent resistor (LDR) as the sensing element. A voltage divider network converts the light levels into proportional analog voltage outputs which feed into one side of the op-amp, while a user defined voltage reference feeds into the opposite side. The op-amp compares the two voltages and provides an output that can be used as a digital signal. If the voltage input fed from the LDR exceeds or is below the predetermined threshold value set by the user, the digital signal output of the opamp activates either an alarm or indicator light through a subsequent electronic driver circuit. The features of the proposed system include low power consumption, simple circuit design, and adjustable sensitivity; these features allow for many different types of applications. Examples of potential uses for this type of system include: security system for intrusion detection, alarm systems activated by darkness or light levels below a certain threshold, monitoring industrial use of light, or as educational tools demonstrating interfacing of analog-signalprocessing-sensor and simple automation.

Index Terms—Ambient Light Sensor, Operational Amplifier Comparator, Light Dependent Resistor (LDR), Threshold-Based Alarm System

I. Introduction

Environmental parameter self-monitoring and control, such as checking ambient light levels, have become an important area in today's industrial, security, and automation sectors. When combined with detection of an event, these parameters can indicate the presence of an unusual event, identify a breach of controlled space or an equipment failure due to excessive ambient temperature variation. All previous examples of light sensor/alarm and monitoring systems are made up of multiple components that either have very complicated designs, draw a lot of current, or lack the flexibility to adapt to changing ambient lighting. Because of this lack of flexibility, most of these systems are not efficient for low-cost, adaptable solutions. This invention provides solutions to the previously identified issues through a light-controlled alarm system using an operational amplifier to operate in a comparator mode. This system uses a light dependent resistor (LDR) to continuously monitor the intensity of the ambient light and convert the optical change into a corresponding electrical output signal. This output signal can then be compared to a pre-defined reference voltage, which can be adjusted, so that when the intensity of light decreases or increases beyond a predetermined threshold value, the system will trigger an alarm or warning signal through a driven output stage. This proposed solution has been designed as a simple circuit and can easily be adjusted to provide increased sensitivity; therefore it is reliable, will provide an immediate response, and uses a small amount of power. As such it can be used for cost-effective applications.

II. Literature Review

A light-intensity controlled alarm system is one kind of alarm system that uses a Light Dependent Resistor (LDR) to detect changes in ambient light. The resistance of an LDR decreases when illuminated. The voltage developed at the LDR's terminals is compared to a fixed reference voltage using a comparator (operational amplifier configured as a comparator). If an increase in ambient light causes the voltage at the LDR terminal to exceed the fixed threshold (set in the past), the output of the comparator sharply transitions from low to high, thereby triggering either a transistor or relay to activate an alarm. An autotransformer regulates the output voltage for stable operation and to reduce instances where voltage varaiations have triggered false alarms. These simple circuits can quickly respond, provide a high level of reliability, and have many security applications including automatic lighting controls and light-based monitoring [1] .

An alarm system based on light intensity control operates on a light dependent resistor (LDR). As the ambient light conditions increase, the resistance of the LDR decreases and increases in the absence of light, therefore this type of device may be used to create light activated switches. The change in resistance creates a voltage corresponding to it which is then compared to a reference voltage set at the op-amp. When the voltage crosses the reference threshold, the output of the op-amp has switched to either maximum or minimum, thus producing a clean, defined and reliable signal to cause the alarm to activate or turn off. To maintain the reliability of the system and reduce false-alarms caused by fluctuation of the power supply, a correctly designed regulated power supply can be used. These systems are inexpensive to manufacture, quick to respond and are suitable for use in many security or automatic control applications [2].

The LDR runs on the principle of measuring the ambient lighting of a specific environment. As solar energy is constantly being issued through windows, the amount of ambient light can fluctuate throughout the day, and this will affect the amount of current flowing through the LDR. When the amount of light that is registered by the LDR exceeds the user's selective threshold, the output voltage from the LDR varies according to the amount of light that it registers. When the output voltage, with respect to a pre-set reference point (the LDR's output voltage) has changed enough to exceed this user's selective threshold, it triggers the electrical device (an alarm) to allow the control circuit to receive the information it requires to operate. Depending upon whether the change in the output voltage occurs above or below the user's selected threshold voltage, the op-amp (pick), will produce either a current flowing down, or a current flowing up, which in turn causes either an alarm activation via a relay or transistors to occur. In order to prevent false alarms and ensure stable performance, proper filtering and regulation are included for the LDR alarm systems. Because of all of these qualities, LDR alarm systems are cost-effective, and are typically used in a

variety of applications, including Home Security, Automatic Lighting and Light Monitoring [3].

A light-intensity controlled alarm system works using an LDR or photo-diode to detect changes in light intensity; the light intensity is then converted into an electrical signal, the level of which corresponds to the amount of light received .An OP-AMP, such as the LM358, was chosen to amplify the signals of all the original transducers. The LM358 was selected due to its low current consumption, wide operating voltage, and ability to operate from a single supply voltage and the good noise immunity. When configured in the comparator mode, the OP-AMP will compare the sensed voltage level to a preset reference level and will provide a high or low output when the sensed voltage goes above or below the reference level. Using the high/low output, either one goes directly to the alarm relay/transistor; filtering and regulating the power supply allows for greater noise immunity thus improving the reliability. Light-based alarm systems are very easy to install, inexpensive, and ideally suited for automatically controlling lights, securing premises, and sensing industrial processes [4].

An alarm system that uses light intensity control uses an LDR (Light Dependent Resistor) to determine the variation of light there by controlling the resistance of the LDR will decrease as the intensity of light hitting it increases. The variation of resistance results in the variation of voltage from voltage divider. The voltage applied to this circuit will vary and this varying voltage signal will be applied to an operational amplifier (such as LM358) that has been set up in the comparator mode to measure the value of the variable voltage input against the value of a preset voltage reference input. This means that when the voltage input from the LDR exceeds that of the reference voltage the output of the comparator will output a signal that has switched states. This switching could be used to energize a transistor or a relay to activate the alarm, etc. The transistor can amplify the current enough to operate high-power devices such as buzzers and lights safely. Filtering and regulated power supplies will help stabilize the device and reduce false triggering. This system is simple and inexpensive to build, responds rapidly to changes in the incoming light intensity and can be used in many different applications including security systems, automatic lighting and industrial sensing [5]

An Alarm System that Controls Light Intensity uses a Sensor to Convert Environmental Condition Changes to Electrical Signals and Amplifies and Compares those signals using an Operational Amplifier configured as a Comparator. The Comparator compares the Sensor's Signal to a predetermined level of Reference Voltage, when this level is exceeded, it creates a fast digital output for reliable switching. The output is sent to a Transistor that operates a Relay, Allowing for the safe operation of an Alarm or Light load. The use of Low-Power Operational Amplifiers enhances the Efficiencies, Stability, and Sensitivity of this system while also reducing the Noise Interference. These systems are Fast, Low Cost to Implement, and Utility Automation for EnergyEfficient and Security Applications [6].

With the Alarm System, you can track light levels with the use of a Light Dependent Resistor (LDR) located in your home or office. An LDR will have decreased resistance = drop off its ability to produce a voltage; hence when the ambient lighting has too high a level of brightness, the LDR will not generate any power so there will be no voltage output from the device. Conversely, when the ambient surroundings have reached a sufficiently low-light level, the LDR will not have enough power to produce an output voltage.

An LDR can be employed to supply sufficient power for operation of an Operational Amplifier (for instance a uA741) when it is used in Comparator mode.The LDR output voltage is compared to a set reference voltage (approximately 6V). Once the LDR output voltage has changed due to either lighting changes or even smoke, the Comparator's output goes from high to low very quickly. The output from the Comparator controls a Transistor and Relay stage that turns on an Alarm or Speaker without overloading the OpAmp. To keep the system running smoothly and to minimize the number of false Alarm, the circuit has a regulated 9V power supply regulated with Filtering Capacitors. The Alarm system has several advantages; it is Fast, Simple, and Reliable for Automatic Alerts and Safety Monitoring [7].

The use of a Light Intensity Control Alarm System utilizes a light sensor (such as an LDR or phototransistor) to sense

changes in the amount of illumination in order to produce an electrical signal that will be proportional to the level of illumination. This electrical signal will then be connected to an operational amplifier configured as a comparator. In continuous operation, the electrical signal will be compared to a reference voltage, which has been predetermined. Once the electrical signal reads above the reference, there will be an instantaneous output from the comparator that will change the output from its support voltage levels to either its minimum or maximum. This process produces a digital control signal that is used by either a transistor or relay as a means to activate the alarm. In addition, with signal conditioning, filtering, and proper amplification, the overall sensitivity of the device will be increased and any interference from noise will be minimized. The Light Intensity Control Alarm System can respond quickly, require a low level of power, and are simple to implement; therefore, they can be effectively used for applications such as monitoring security, controlling and monitoring the control of lighting, and for industrial safety [8] An alarm system that detects the absence of light and responds to light changes automatically. This system uses a device called a light dependent resistor (also known as an LDR or light sensor), which changes its resistance based on how much light is present. To determine how much light is present in an area, an LDR is connected to an electrical circuit using a standard resistor as a voltage divider (this means that the voltage across the LDR can be measured). The voltage generated by this voltage divider is sent to an operational amplifier configured in comparator mode to compare the voltage output from the LDR to a set reference voltage. When the voltage from the LDR falls below this reference voltage, the operational amplifier switches from one output state (high) to another output state (low). The output of the operational amplifier can be increased further by using either a relay or transistor to safely turn on or off the alarm load at a higher level of voltage. Filtering and using a regulated power supply will improve the overall stability of the pdf to further limit false alarms. This method of operation provides fast response time, simplicity in design, and reliability when used for security purposes, such as in automatic monitoring systems
[9].

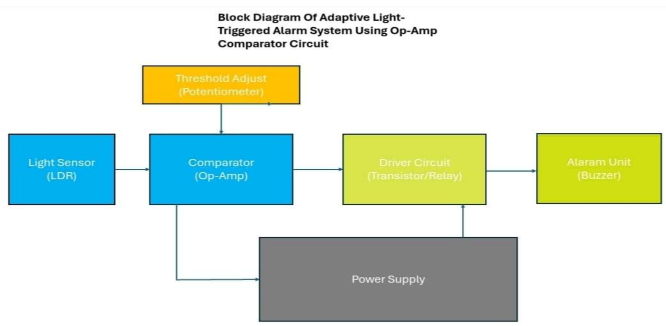
A light intensity-control alarm system uses a lightdependent resistor (LDR) as a way to detect changes in illumination because an LDR's resistance decreases when exposed to a higher amount of light. This change in resistance creates a corresponding change in voltage from the LDR (via a potential divider circuit). The voltage created by the LDR is passed into an operational amplifier configured to operate as a comparator. The LDR voltage is compared by the comparator with a predetermined reference voltage. If the voltage from the LDR exceeds (or drops below) the determined level for that predetermined threshold, the result is that the output of the comparator "switches" quickly to create a digital output that can be applied to a transistor or relay to turn on the alarm. The LDR's output can also be effectively conditioned through proper filtering and regulated power supply to increase sensitivity, stability, and noise immunity. Due to their speed of operation, simple design, and low power consumption, this type of alarm system can be used for various security monitoring purposes, as well as to automatically control lights [10].

III. Methodology

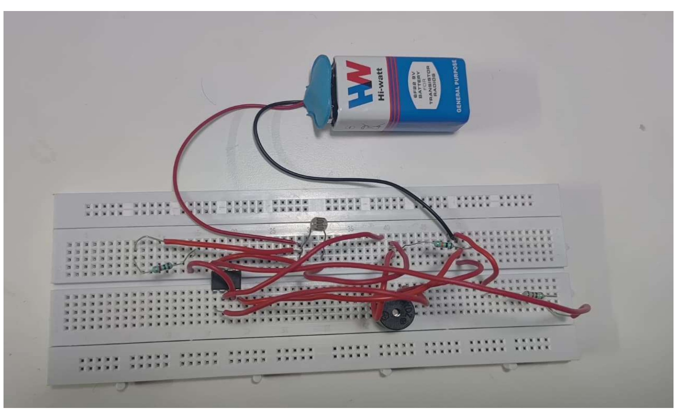
The most important part of this project is seeing through to a Light Dependent Resistor (LDR) and an LM358 OPAMP, whose function will be to do a comparison of the light level coming from the LDR against the voltage set for the project. The project will use LDR as the light intensity sensing element. An LDR will be placed in series with a 10 k resistor to form a circuit for the measurement of natural light in the surroundings. The electrical resistance of the LDR will decrease with an increase in light and vice versa; hence, the corresponding voltage output of the voltage divider will change, thereby giving an electrical representation of the intensity of the ambient light present. In this case, the voltage reference is supplied by a preset potentiometer of 10 k. The alarm activation threshold is determined by this reference voltage which is based on the intensity of the alarm. The potentiometer provides the user an option to set the desired sensitivity of the system, hence making the circuit adaptable to different lighting environments by just one adjustment of the potentiometer. The LM358 is an operational amplifier that goes into comparator mode for this application. The LDR voltage divider supplies the non-inverting input (+) of the opamp while the reference voltage from the potentiometer goes to the inverting input (–). When the voltage from the LDR is greater than or has dropped below the reference level, the output of the op-amp toggles between high and low thus the op-amp behaves as a comparator. The op-amp output is then used to turn on an LED through a current limiting resistor as a visual indication of the alarm condition. The output from the opamp also drives a 5V DC buzzer for an audible indication.

IV. Result and Discussion

A Light-Intensity Controlled Alarm Circuit using Op-Amps in a comparator mode has been built and has functioned properly and tested correctly with respect to the independent variable of this experiment: the intensity of light in the environment. The observation made during the experiment indicated that the resistance of the Light Dependent Resistor (LDR) was variable with the change in light intensity. In other words, an increase or decrease in the surrounding light would also result in an increase or decrease of the input voltage to the Op-Amp. Conclusion will be reached after verifying that the LDR appropriately detects light. When potentiometer readings indicate excessive light, the Op-Amp operates as a comparator, and will provide a continuously variable output based on how much light was detected.When more light is detected by the Op-Amp, it causes the output of the Op-Amp to increase. This is seen in the LED lighting up and the buzzer sounding when an increase in light is detected by the system and an alarm alert has been sounded. Once the amount of light falls below a preset normal level defined as the preset norm, the output of the Op-Amp will fall to the lowest possible level and both LED and buzzer will go off, thus confirming that the system worked correctly as intended when light levels were rising and falling. An important note on the implementation of the transistor as an amplifying device for the Buzzer; this had a positive outcome, as it allowed the Buzzer to have a louder output without running the Op-Amp into an overdriven condition. The Operation of the Circuit was safe and provided a smooth operation with a 9V battery supply. This data demonstrates that the Project can adequately detect changes in ambient light levels and activate the Alarm.



(a) Block diagram of Adaptive Light-Triggered Alarm System Using Op-Amp Comparator Circuit



(b) Final image output of the model

V. Conclusion

In conclusion, this research was done on a light monitoring alert system to control the level of illumination using an operational operational amplifier with a comparator layout. This system was completely constructed and operated efficiently for an ambient light level measuring device using an LDR and comparing the measured signal to a pre-established reference level of illumination. At any point in time, any increase or decrease from the pre-set reference point was indicated by activating an LED and buzzer to provide visual and audible signals when the change in the amount of illumination was exceeded. The total design cost was quite economical because all circuit components were readily available and common items such as LM358, resistors, a potentiometer, and a buzzer were used as part of the circuit structure. In addition, an adjustment to the potentiometer permitted a range of control regarding the system's sensitivity which provided flexibility to address a variety of ambient light levels. The experimental data produced from the project work conducted demonstrated the results of the project and the validity of the use of a comparator to address the issue presented. This project serves as an example of an actual application of op-amps utilized in a real-time environmental monitoring application. The circuit's ease of design and reliability in operation will allow for further application as a method for maintaining basic monitoring/security control of locations in which the detection of light can occur.

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